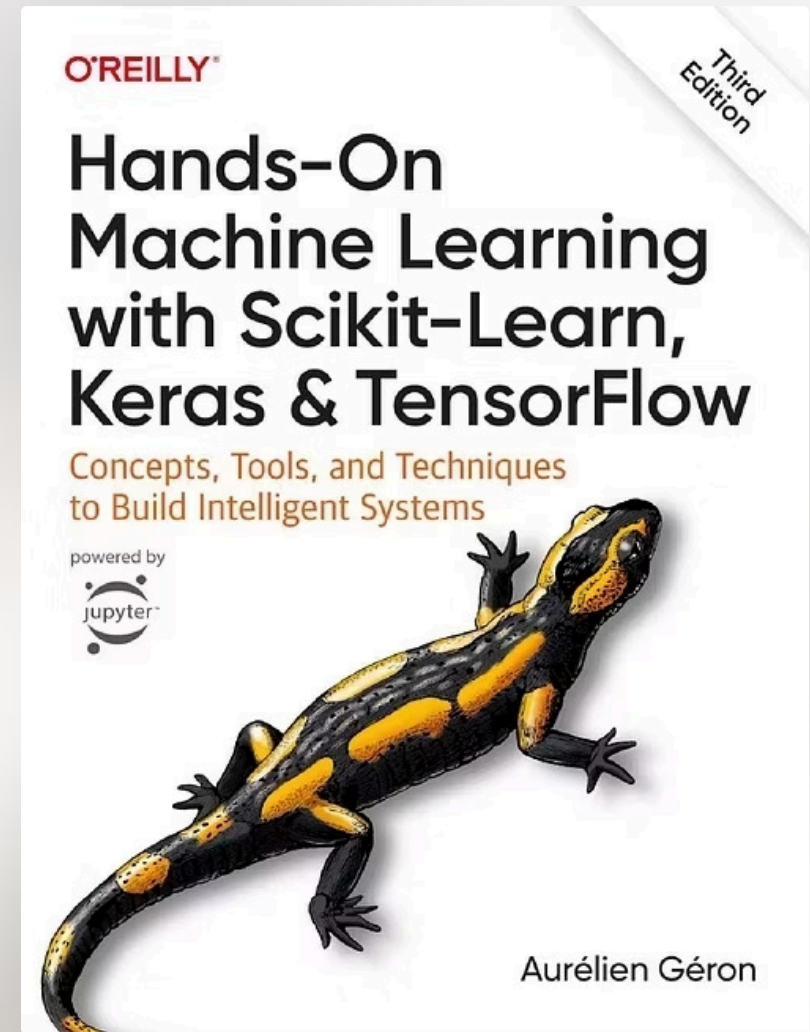


Machine Learning

Lessons based on the Textbook "Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow" by Aurélien Géron, 2023.

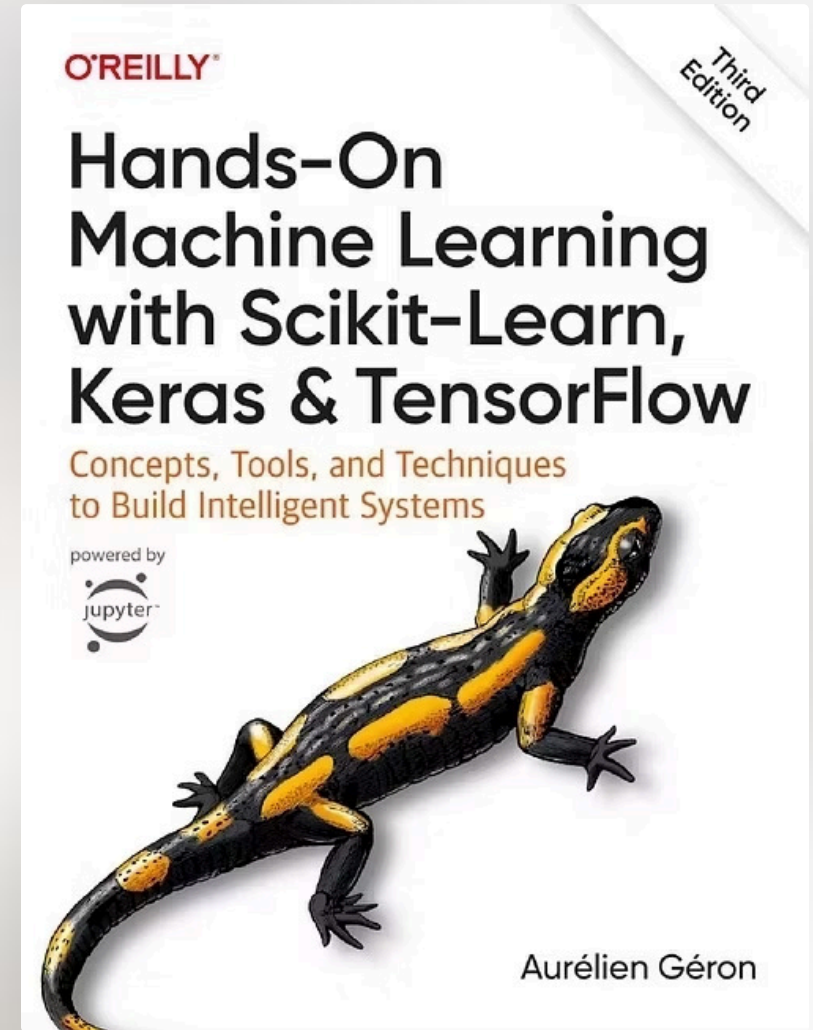
RP by Rodrigo Martins Pagliares

Last update: 07/01/2025



Machine Learning

Chapter 1 – The Machine Learning Landscape





Agenda for Chapter 1

1. **What Is Machine Learning?**
2. Why Use Machine Learning?
3. Examples of Applications
4. Types of Machine Learning Systems
5. Main Challenges of Machine Learning
6. Testing and Validating

Agenda for Chapter 1

1. What Is Machine Learning?

2. Why Use Machine Learning?

3. Examples of Applications

4. Types of Machine Learning Systems

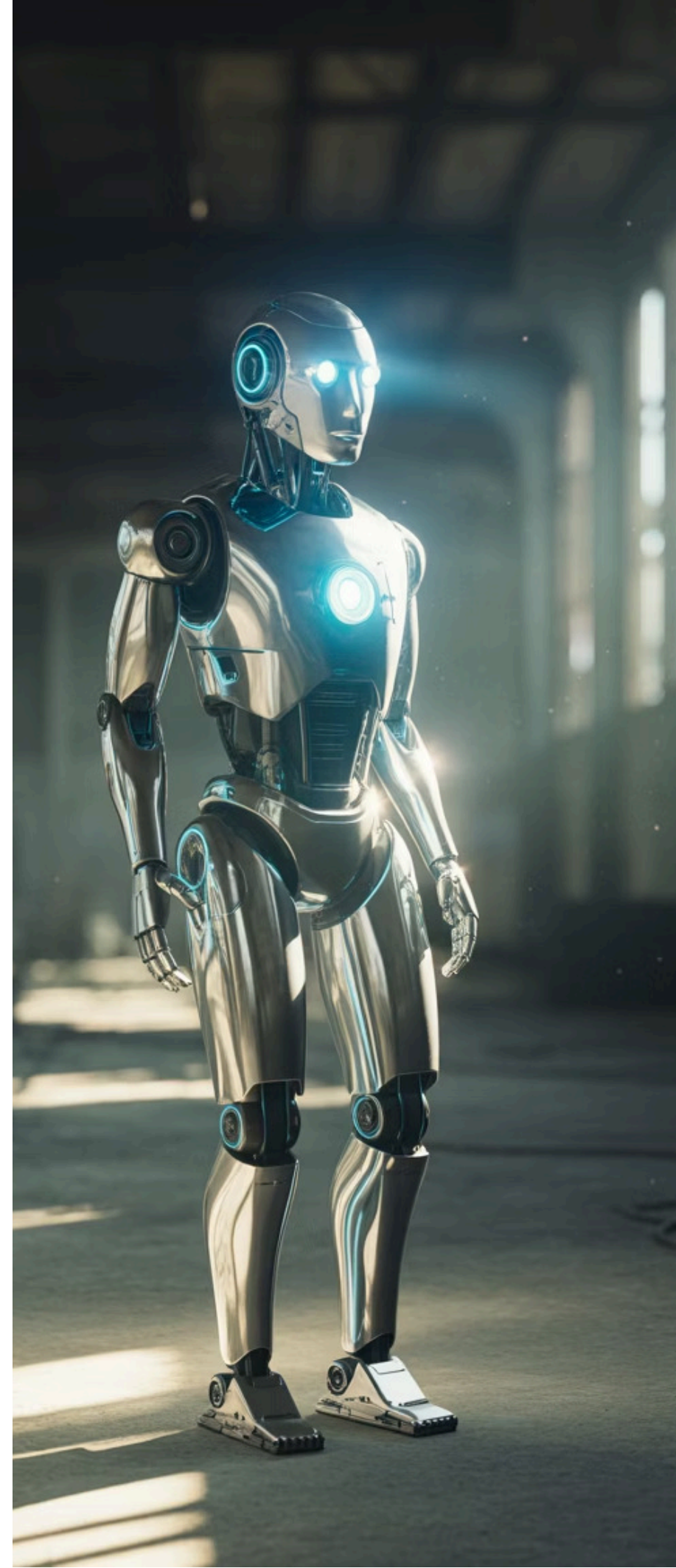
- Classification criteria #1: Supervision during training
 - Supervised, Unsupervised, Semi-Supervised, Self-Supervised, and Reinforcement Learning
- Classification criteria #2: Approach to Learning Method
 - Batch and Online (Incremental) Learning
- Classification criteria #3: Learning Method
 - Instance-Based and Model-Based Learning

5. Main Challenges of Machine Learning

- Insufficient Quantity of Training Data
- Non-representative Training Data
- Poor-Quality Data
- Irrelevant Features
- Overfitting the Training Data
- Underfitting the Training Data
- Stepping Back

6. Testing and Validating

- Hyperparameter Tuning and Model Selection
- Data Mismatch



What Is Machine Learning?

What Is Machine Learning?

Arthur Samuel's Definition (1959)

"Machine learning is the field of study that gives computers the ability to learn without being explicitly programmed."

Tom Mitchell's Definition (1997)

"A computer program **learns from experience** (E) with respect to **task** (T) and **performance measure** (P), if its **performance on T improves with experience E.**"

Aurélien Géron's Hands-On book

"Machine learning is the **science (and art)** of **programming** computers so they can **learn from data.**"

Example

Spam filters learn to recognize and flag spam emails by using examples of labeled data (spam/nospam - a.k.a "ham").



Training Set, Training Instance, and Model

The **examples** that the **system** uses to **learn** are called the **training set**.

Each **training example** is called a **training instance** (or **sample**).

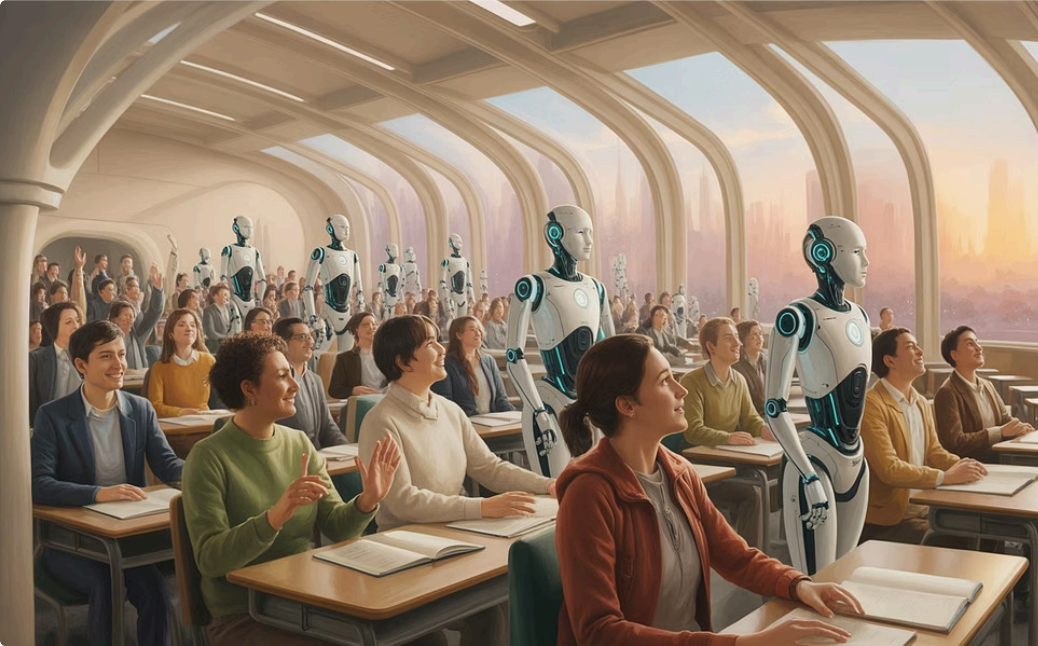
The part of a **ML system** that **learns** and **makes predictions** is called a **model**.





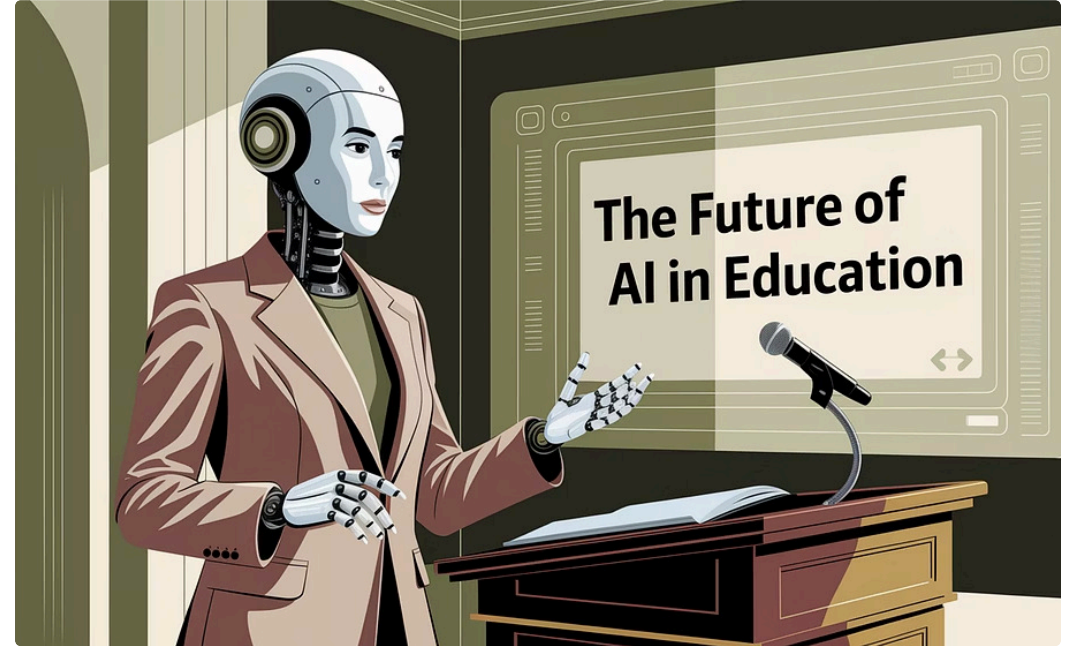
**Next: Why use
Machine Learning?**

Q&A



Open Floor

Open floor for discussion or clarifications



Discussion

Share your thoughts and questions about AI concepts